

QUBO Formulation: HP Lattice Protein Conformation Selection

HP Lattice Protein Conformation Selection

Problem Statement. Given a hydrophobic-polar (HP) protein sequence and a curated set of N candidate lattice conformations, each conformation $i \in \{0, \dots, N-1\}$ is associated with a contact energy $E_i < 0$ that quantifies the stability contribution of nonconsecutive hydrophobic-hydrophobic contacts. The task is to select exactly one conformation such that the total contact energy is minimized.

Decision Variables. For each candidate conformation $i \in \{0, \dots, N-1\}$, introduce a binary decision variable $x_i \in \{0, 1\}$. The variable $x_i = 1$ indicates that conformation i is selected, and $x_i = 0$ otherwise.

QUBO Derivation. Step 1 — Original Objective. We wish to minimize the total contact energy over the binary decision variables:

$$\min_{x \in \{0,1\}^N} \sum_{i=0}^{N-1} E_i x_i.$$

Step 2 — Constraint. Exactly one conformation must be selected, which is expressed as the equality constraint:

$$\sum_{i=0}^{N-1} x_i = 1.$$

Step 3 — Penalty Term Expansion. We enforce the equality constraint by adding a quadratic penalty term to the objective. Let $\lambda > 0$ be the penalty weight. The penalty term is written as:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} x_i - 1 \right)^2.$$

Expanding the square and applying the idempotent property $x_i^2 = x_i$ for binary variables

yields:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} x_i^2 + \sum_{i \neq j} x_i x_j - 2 \sum_{i=0}^{N-1} x_i \right) \quad (1)$$

$$= \lambda \left(\sum_{i=0}^{N-1} x_i + 2 \sum_{0 \leq i < j \leq N-1} x_i x_j - 2 \sum_{i=0}^{N-1} x_i \right) \quad (2)$$

$$= -\lambda \sum_{i=0}^{N-1} x_i + 2\lambda \sum_{0 \leq i < j \leq N-1} x_i x_j + \lambda. \quad (3)$$

The resulting general penalty contribution to the Hamiltonian consists of linear terms with coefficient $-\lambda$, quadratic terms with coefficient 2λ , and a constant offset λ .

Step 4 — Combined Hamiltonian. The full QUBO Hamiltonian $H(x)$ is the sum of the original objective and the penalty term:

$$H(x) = \sum_{i=0}^{N-1} E_i x_i - \lambda \sum_{i=0}^{N-1} x_i + 2\lambda \sum_{0 \leq i < j \leq N-1} x_i x_j + \lambda \quad (4)$$

$$= \sum_{i=0}^{N-1} (E_i - \lambda) x_i + 2\lambda \sum_{0 \leq i < j \leq N-1} x_i x_j + \lambda. \quad (5)$$

Step 5 — Q Matrix Entries and Offset. Reading off the coefficients from the general closed form $H(x) = x^\top Q x + c$, we obtain:

$$Q_{i,i} = E_i - \lambda, \quad \text{for all } i \in \{0, \dots, N-1\}, \quad (6)$$

$$Q_{i,j} = 2\lambda, \quad \text{for all } 0 \leq i < j \leq N-1, \quad (7)$$

$$Q_{j,i} = 2\lambda, \quad \text{for all } 0 \leq i < j \leq N-1, \quad (8)$$

$$c = \lambda. \quad (9)$$

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any violation of the equality constraint incurs a higher energy cost than the maximum possible reduction in the objective function. The maximum possible decrease in the objective function by selecting a different conformation is bounded by $\max_i |E_i|$. Therefore, setting $\lambda > \max_i |E_i|$ guarantees that any infeasible assignment (where $\sum_i x_i \neq 1$) yields a penalty of at least λ , which strictly dominates any feasible objective improvement. This ensures the global minimum of $H(x)$ over $\{0, 1\}^N$ corresponds to a feasible solution.

Correctness Argument. Minimizing $H(x)$ over binary vectors recovers the optimal conformation selection because (i) for any feasible solution satisfying $\sum_i x_i = 1$, the penalty term vanishes and $H(x)$ reduces exactly to the original objective $\sum_i E_i x_i + \lambda$; (ii) for any infeasible solution where $\sum_i x_i \neq 1$, the penalty term evaluates to at least λ . Since $\lambda > \max_i |E_i|$, the energy of any infeasible solution strictly exceeds the energy of the best feasible solution. Consequently, the global minimizer of $H(x)$ must be feasible and, by construction, minimizes the original contact energy.

Illustrative Example. Consider a small instance with $N = 3$ candidate conformations and contact energies $E_0 = -5$, $E_1 = -3$, and $E_2 = -4$. We choose $\lambda = 6$, which satisfies $\lambda > \max\{|E_0|, |E_1|, |E_2|\} = 5$. Substituting these values into the general formulas yields the concrete Hamiltonian:

$$H(x) = (-5 - 6)x_0 + (-3 - 6)x_1 + (-4 - 6)x_2 + 12(x_0x_1 + x_0x_2 + x_1x_2) + 6 \quad (10)$$

$$= -11x_0 - 9x_1 - 10x_2 + 12x_0x_1 + 12x_0x_2 + 12x_1x_2 + 6. \quad (11)$$

The corresponding Q matrix entries and offset are:

$$Q_{0,0} = -11, \quad Q_{1,1} = -9, \quad Q_{2,2} = -10, \quad (12)$$

$$Q_{0,1} = Q_{1,0} = Q_{0,2} = Q_{2,0} = Q_{1,2} = Q_{2,1} = 12, \quad (13)$$

$$c = 6. \quad (14)$$

Enumerating all $2^3 = 8$ binary assignments, the feasible solutions are $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$. Their energies are $H(1, 0, 0) = -11 + 6 = -5$, $H(0, 1, 0) = -9 + 6 = -3$, and $H(0, 0, 1) = -10 + 6 = -4$. The infeasible solutions all incur a penalty of at least 6, yielding energies strictly greater than -5 . Thus, the global minimum is attained at $x^* = (1, 0, 0)$ with minimum energy -5 , correctly selecting conformation 0.